

T(4th Sm.)-Computer Science-H/CC-9/CBCS

2021

COMPUTER SCIENCE — HONOURS

Paper : CC-9

(Introduction to Algorithms and Its Application)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** from the rest.

1. Answer **any five** questions :

2×5

- (a) Mention the time complexity of BFS traversal of a graph.
- (b) Define Ω . Show that $10n^2 + 6n + 3$ is $\Omega(n^2)$.
- (c) What do you understand by worst case time complexity of an algorithm? Give an example.
- (d) Define P and NP class of problems.
- (e) Differentiate recursive and non-recursive algorithm.
- (f) Briefly state an experiment where divide-and-conquer rule approach is suitable.
- (g) Mention one advantage and one disadvantage of Greedy algorithm.
- (h) Define minimum spanning tree.

2. (a) Mention the purpose of Floyd-Warshall algorithm.

(b) Write down Floyd-Warshall algorithm.

2+8

3. (a) Briefly discuss about Travelling Salesman Problem.

(b) Solve the following Travelling Salesman Problem (TSP) using dynamic approach.

There are four cities 1, 2, 3 and 4. Start from city 1 visit all cities.

The cost matrix is given below :

3+7

	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

Please Turn Over

4. (a) Differentiate between divide-and-conquer methods and Dynamic programming approach.
 (b) Apply dynamic programming approach to find the following matrix chain multiplication.

$A_1 A_2 A_3 A_4 A_5 A_6$

<u>Matrix</u>	<u>Dimension</u>
A_1	30×35
A_2	35×15
A_3	15×5
A_4	5×10
A_5	10×20
A_6	20×25

3+7

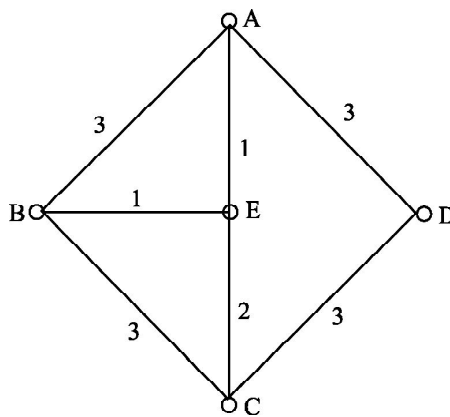
5. (a) Briefly state the graph colouring problem.

- (b) Write down the BFS algorithm.

4+6

6. (a) Write down Kruskal's algorithm.

- (b) Find the Kruskal's algorithm, a minimal spanning tree of the weighted graph.



5+5

7. (a) Explain briefly the strategy used in divide-and-conquer (D & C) method.

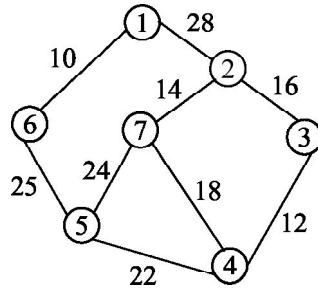
- (b) Write the algorithm of D & C method for a problem P. State the recurrence relation for computing time of D & C method.

4+4+2

(3)

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8. (a) State the Knapsack problem formally. What is its time complexity using Greedy algorithm?
(b) Apply Prim's algorithm to find the minimum spanning tree of the graph given below.



(3+2)+5
